MARINE WILDLIFE

RETURN OF THE OLIVE RIDLEYS

By Arkamoy Datta Majumdar



SPECTACLE OF 'ARRIBADA'
Thousands of Olive Ridley turtles crawl out from the sea to lay eggs in Odisha's Rushikulya beach

First came the shock and dismay—throughout January, the discovery of hundreds of carcasses of Olive Ridley turtles on the beaches in and around Chennai raised the hackles of wildlife conservationists. Then, from mid-February and through March, emerged a remarkable silver lining: the Rushikulya beach in Odisha's Ganjam district witnessed an unprecedented mass nesting event, with close to 700,000 turtles laying eggs. This astonishing spectacle comes after the site saw no nesting at all last year, making it a moment of triumph for conservationists. Redemption, it seemed, had arrived further up the coast.

Olive Ridley turtles (*Lepidochelys olivacea*) are the most abundant of all sea turtles and they inhabit the warm waters of the Pacific, Atlantic and Indian Oceans. Olive Ridleys travel 9,000 km from the Pacific Ocean, as well as shorter distances from the Indian Ocean, to breed and nest on select Indian beaches in Tamil Nadu, Andhra Pradesh and Odisha, which is home to the two mass-nesting beaches for them. Olive Ridleys are a fundamental link in marine ecosystems by regulating populations of organisms like jellyfish and helping maintain coral reefs. But they face numerous threats, including the mass harvesting of eggs, poaching and unintended capture in fishing nets, leading to injuries or drowning.

As thousands of these marine reptiles crawled ashore beneath the moonlit sky—and, defying age-old practice, during daylight hours as well—locals and conservationists stood spellbound. This extraordinary phenomenon, known as 'arribada' (Spanish for 'arrival'), has surpassed previous records. The sheer scale of this nesting event stands as a testament to years of painstaking conservation efforts, yet the very rarity of this moment of success underscores the ongoing challenges posed by climate change and habitat disruption.

Gahirmatha beach in Kendrapara district, till now the world's largest known nesting ground for Olive Ridleys, has recorded over 600,000 turtles this year. However, unlike Rushikulya, the site has never seen a drastic decline in numbers. The mass nesting at Rushikulya began on February 16 and continued for eight days, with the tally of nesting turtles touching 698,718. Gahirmatha recorded 606,933 turtles between March 5 and 9. Olive Ridleys begin their mating rituals in the sea around September-October, and the nesting season lasts from January to March.

Experts attribute the historic nesting event at Rushikulya to a combination of favourable climatic conditions and proactive conservation management strategies. Dibya Sankar Behera, assistant conservator of forests and in-charge of the Olive Ridley (Sea Turtle) Protection Area in Odisha, is delighted. "All our efforts—strict fishing restrictions, consistent beach clean-ups and awareness campaigns—played a role. The absence of rain ensured that the beach remained dry, providing the perfect nesting environment," he says.

Local involvement has also been a crucial factor. Rabindranath Sahu, a member of the Rushikulya Sea Turtle Protection Committee, says, "The community once saw turtles as a nuisance. Now, they take pride in cleaning the beach and ensuring a safe nesting environment for the turtles," he says.

A key highlight of this year's nesting at Gahirmatha Marine Sanctuary was the return of turtles—300,000 in total—to Ekakulanasi Island after 33 years. The beach had suffered significant sea erosion, leading to a reduction in its profile. However, since 2020, the beach has been undergoing natural accretion, causing it to extend once more.

OLIVE RIDLEYS KEPT OFF RUSHIKULYA LAST YEAR, BUT NEARLY 700,000 HAVE NESTED ON THE BEACH IN 2025. BETTER CLIMATE AND CONSERVATION MANAGEMENT STRATEGIES HELPED IN THE STUNNING TURNAROUND